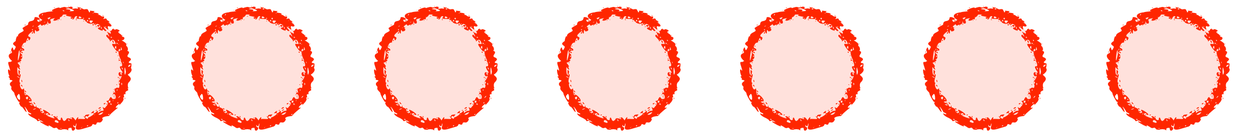
Advanced Algorithms

Hashing

刘明谋 Nanjing University, Suzhou, 2026

Balls into Bins





Balls into Bins (Random Function)

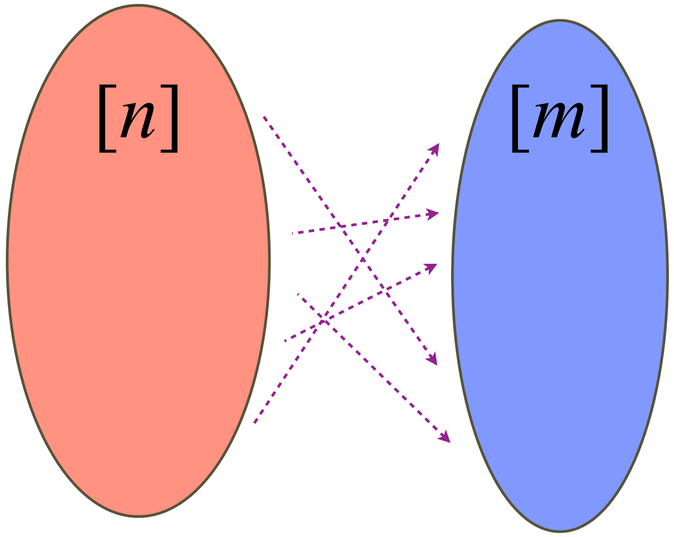
- n balls into m bins:

$$\Pr[\text{assignment}] = \frac{1}{m} \cdots \frac{1}{m} = \frac{1}{m^n}$$

- uniform random function:

$$\Pr[f] = \frac{1}{\left|[n] \to [m]\right|} = \frac{1}{m^n}$$

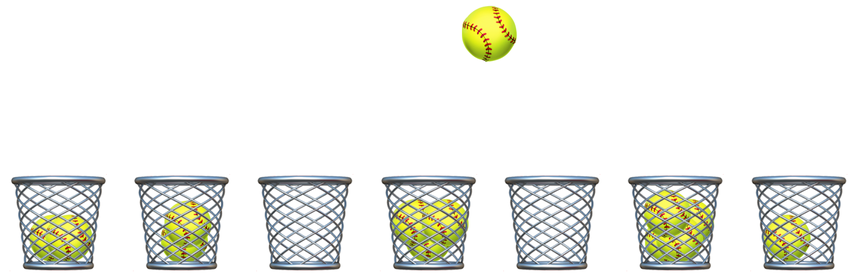
1-1	birthday
on-to	coupon collector
pre-image size	occupancy



uniform random function

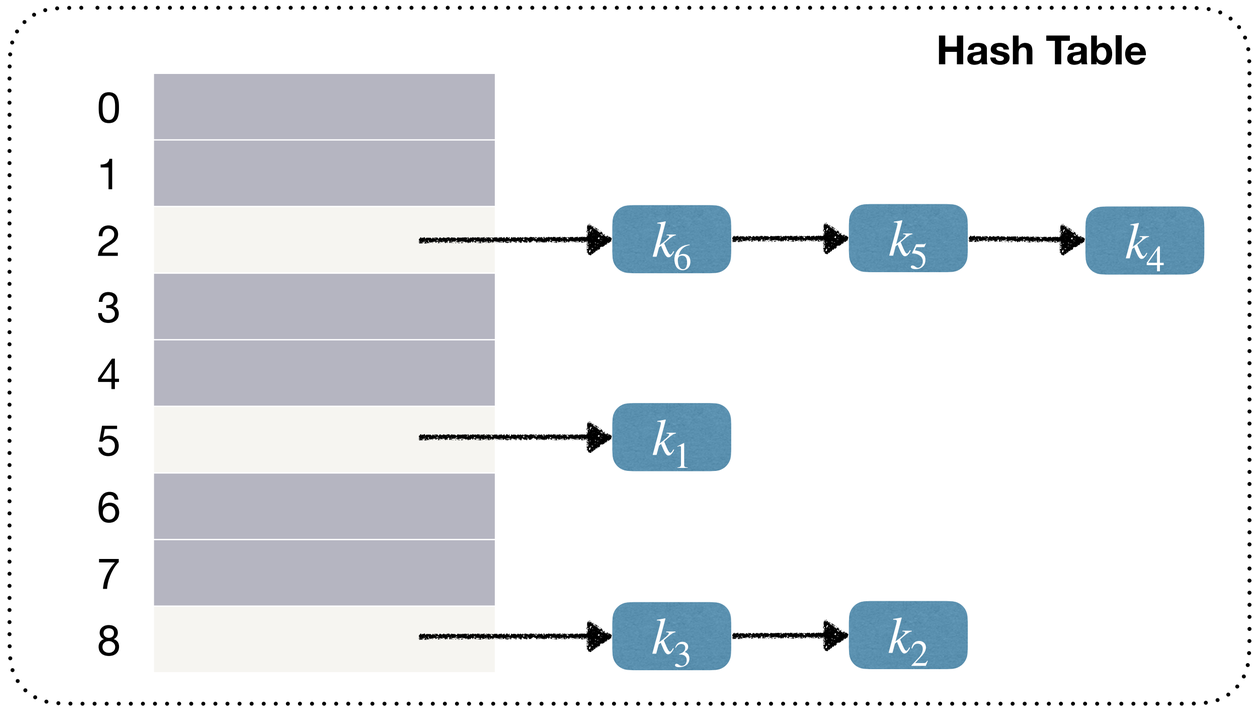
$$f : [n] \to [m]$$

Load Balancing



Chaining

- Each bucket i stores a pointer to a linked list L_i.
- All keys that are hashed to index i go to L_i.



$h(k_4) = h(k_5) = h(k_6) = 2$

Space Cost:

- $\Theta(m)$ for pointers;
- $\Theta(n)$ for actual elements.

Hashing with Chaining

- `Search(k)` where k is a key. ---------- Time depends on length of the linked list!

  Search can cost $\Theta(n)$ in worst-case. (All keys hash to same value.)

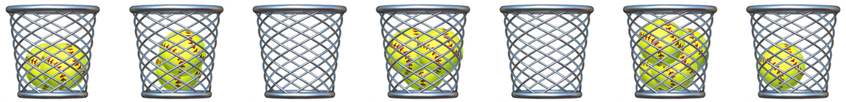
  - Compute $h(k)$
  - Go through the corresponding list to search item with key k.

- `Insert(x)` where x is a pointer to an item. ---------- $O(1)$ Note: we assume computing h takes $O(1)$ time.
  - Compute $h(x.key)$
  - Insert x to the head of the corresponding list.

- `Remove(x)` where x is a pointer to an item. ---------- $O(1)$
  - Simply remove x from the linked list.

Balls into Bins

(Multinomial distribution)



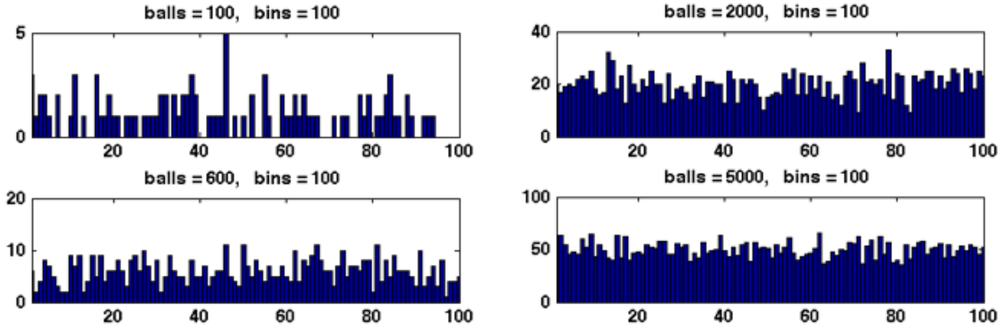
- Throw m balls into n bins ***u.a.r.*** Numbers of balls received in each bins:

$$(X_1, X_2, \ldots, X_n) \sim \textbf{multinomial distribution of parameters } m, (\underbrace{1/n, \ldots, 1/n}_{n})$$

- <u>Occupancy problem</u>: the maximum load $\max_{1 \le i \le n} X_i$ ***w.h.p.*** (*with high probability*)

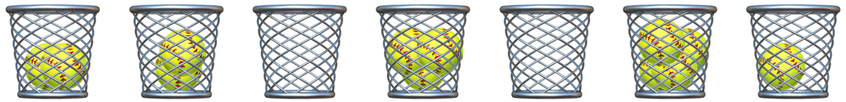
$$\max_{1 \le i \le n} X_i = \begin{cases} O\left(\dfrac{\log n}{\log \log n}\right) & \text{if } m = n \\ O\left(\dfrac{m}{n}\right) & \text{if } m \ge n \ln n \end{cases}$$

w.h.p. (with prob. $1 - o\left(\frac{1}{n}\right)$)



Balls into Bins

(Multinomial distribution)

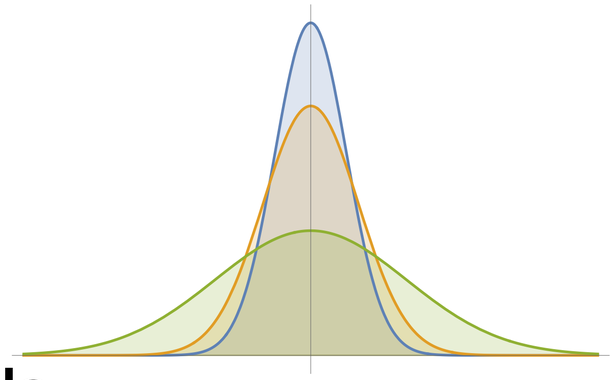


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$$(X_1, X_2, \ldots, X_n) \sim \textbf{multinomial distribution of parameters } m, (1/n, \ldots, 1/n)$$

- <u>Occupancy problem</u>: the maximum load $\max_{1 \le i \le n} X_i$ ***w.h.p.*** (***with high probability***)
- Markov: $\mathbb{E}[X_i] = 1$, $\Pr[X_i \ge k] \le 1/k$. Union bound: $\Pr[\max_i X_i \ge k] \le n/k$.
- Chebyshev: $X_i \sim \mathrm{Bin}(n, 1/n)$. $\mathrm{Var}(X_i) \le 1$. $\Pr[\,|X_i - 1| \ge \sqrt{n}\,] \le 1/n$.
- Binomial distribution: $X_i \sim \mathrm{Bin}(n, 1/n)$, $\Pr[X_i \ge k] = \sum_{j \ge k} \binom{n}{j} \left(\frac{1}{n}\right)^j \le \binom{n}{k} \left(\frac{1}{n}\right)^k$
  - **Fact**: $\binom{n}{k} \le \left(\frac{ne}{k}\right)^k$ $\binom{n}{k} \left(\frac{1}{n}\right)^k \le \left(\frac{e}{k}\right)^k \le \frac{1}{n^{O(1)}}$ if $k = \Omega(\log n / \log\log n)$.

Chernoff Bound



- Chernoff bound: Let $X_1, \ldots, X_n \in \{0,1\}$ be **independent trials**

$$X = \sum_{i=1}^{n} X_i \quad \text{and} \quad \mu = \mathbb{E}[X]$$

(Poisson binomial RV with mean μ)

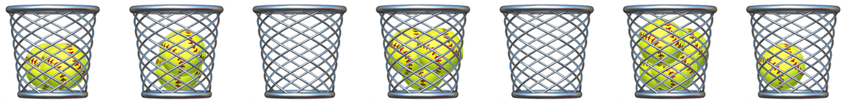
- For any $\delta > 0$,

$$\Pr(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{(1+\delta)}} \right)^{\mu}$$

- For any $0 < \delta < 1$,

$$\Pr(X \leq (1-\delta)\mu) \leq \left(\frac{e^{-\delta}}{(1-\delta)^{(1-\delta)}} \right)^{\mu}$$

Occupancy Problem



- Throw m balls into n bins ***u.a.r.*** The i-th bin receives X_i balls:

$$\text{Marginally,} \quad X_i \sim \text{Bin}(m, 1/n) \quad \text{and} \quad \mu = \mathbb{E}\left[X_i\right] = \frac{m}{n}$$

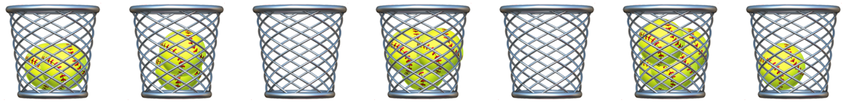
- When $m = n$: $\mu = 1$

$$\Pr\left(X_i \geq L\right) = \Pr\left(X_i \geq L\mu\right) \leq \frac{\mathrm{e}^L}{\mathrm{e}L^L} \leq \frac{1}{n^2} \qquad \text{for } L = \frac{\mathrm{e}\ln n}{\ln\ln n}$$

Chernoff: $\Pr(X_i \geq (1+\delta)\mu) \leq \left(\frac{\mathrm{e}^\delta}{(1+\delta)^{(1+\delta)}}\right)^\mu$

- **Union bound:** $\Pr\left(\max_{1\leq i\leq n} X_i \geq L\right) \leq \sum_{i=1}^{n} \Pr\left(X_i \geq L\right) \leq \frac{1}{n} \implies \max_{1\leq i\leq n} X_i = O\left(\frac{\log n}{\log\log n}\right)$ **w.h.p.**

Occupancy Problem



- Throw m balls into n bins ***u.a.r.*** The i-th bin receives X_i balls:

$$\text{Marginally,} \quad X_i \sim \text{Bin}(m, 1/n) \quad \text{and} \quad \mu = \mathbb{E}\left[X_i\right] = \frac{m}{n}$$

- When $m \geq n \ln n$: $\mu \geq \ln n$

$$\Pr\left(X_i \geq \frac{2\mathrm{e}m}{n}\right) = \Pr\left(X_i \geq 2\mathrm{e}\mu\right) \leq 2^{-2\mathrm{e}\mu} \leq 2^{-2\mathrm{e}\ln n} \leq \frac{1}{n^2}$$

Chernoff: $\Pr\left(X_i \geq L\right) \leq 2^{-L}$ if $L \geq 2\mathrm{e}\mu$

- **Union bound**: $\Pr\left(\max_{1\leq i\leq n} X_i \geq \frac{2\mathrm{e}m}{n}\right) \leq \sum_{i=1}^{n} \Pr\left(X_i \geq \frac{2\mathrm{e}m}{n}\right) \leq \frac{1}{n} \implies \max_{1\leq i\leq n} X_i = O\left(\frac{m}{n}\right)$ **w.h.p.**

Chernoff Bound (Upper Tail)

- Let $X_1, \ldots, X_n \in \{0,1\}$ be independent trials with $X = \sum_{i=1}^{n} X_i$ and $\mu = \mathbb{E}[X]$

$$\text{For any } \delta > 0: \quad \Pr(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{(1+\delta)}} \right)^{\mu}$$

Proof: $\Pr(X \geq (1+\delta)\mu) \leq \Pr\left(e^{tX} \geq e^{t(1+\delta)\mu}\right)$ (for any $t > 0$)

(Markov's inequality) $\leq e^{-t(1+\delta)\mu} \cdot \mathbb{E}\left[e^{tX}\right]$

Moment Generating Function

- The moment generating function (MGF) of a random variable X is

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_{k \geq 0} \frac{t^k \mathbb{E}[X^k]}{k!}$$

- If $X_1, \ldots, X_n \in \{0,1\}$ are independent with $X = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[X]$, then

$$M_X(t) = \mathbb{E}[e^{tX}] \leq e^{(e^t - 1)\mu}$$

Proof: $\mathbb{E}\left[e^{tX}\right] = \mathbb{E}\left[\prod_{i=1}^n e^{tX_i}\right] = \prod_{i=1}^n \mathbb{E}\left[e^{tX_i}\right] = \prod_{i=1}^n \left(e^t p_i + (1 - p_i)\right) \leq \prod_{i=1}^n e^{(e^t - 1)p_i}$

(independence) (where $p_i = \mathbb{E}[X_i]$) (because $1 + x \leq e^x$)

$= e^{\sum_{i=1}^n (e^t - 1)p_i} = e^{(e^t - 1)\mu}$ (where $\mu = \Sigma_{i=1}^n p_i$)

Chernoff Bound (Upper Tail)

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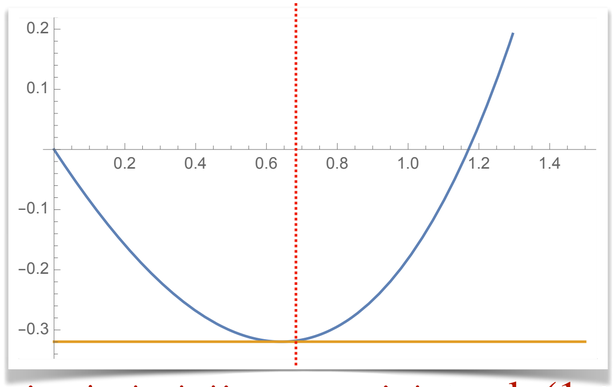
For any $\delta > 0$: $\Pr(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{(1+\delta)}}\right)^{\mu}$

Proof: $\Pr(X \geq (1+\delta)\mu) \leq \Pr\left(e^{tX} \geq e^{t(1+\delta)\mu}\right)$ (for any $t > 0$)

(Markov's inequality) $\leq e^{-t(1+\delta)\mu} \cdot \mathbb{E}\left[e^{tX}\right] \leq e^{-t(1+\delta)\mu} \cdot e^{(e^t-1)\mu}$

$= e^{(e^t-1-t(1+\delta))\mu}$ $= \left(\frac{e^{\delta}}{(1+\delta)^{(1+\delta)}}\right)^{\mu}$

(choose $t = \ln(1+\delta)$)



(minimized at stationary point $t = \ln(1+\delta)$)

Chernoff Bound (Lower Tail)

- Let $X_1, \ldots, X_n \in \{0,1\}$ be independent trials with $X = \sum_{i=1}^{n} X_i$ and $\mu = \mathbb{E}[X]$

For any $0 < \delta < 1$: $\Pr(X \leq (1-\delta)\mu) \leq \left(\frac{e^{-\delta}}{(1-\delta)^{(1-\delta)}}\right)^{\mu}$

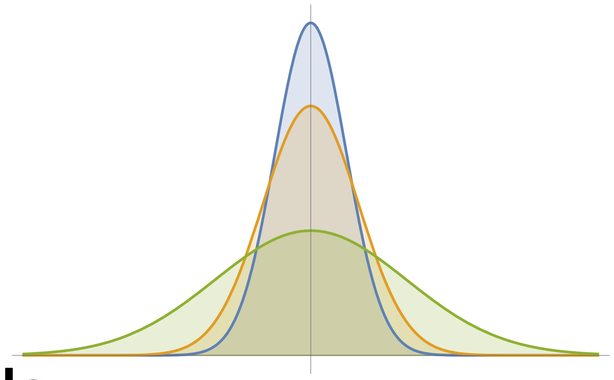
Proof: $\Pr(X \leq (1-\delta)\mu) \leq \Pr\left(e^{tX} \geq e^{t(1-\delta)\mu}\right)$ (for any $t < 0$)

(Markov's inequality) $\leq e^{-t(1-\delta)\mu} \cdot \mathbb{E}\left[e^{tX}\right] \leq e^{-t(1-\delta)\mu} \cdot e^{(e^t-1)\mu}$

$= e^{(e^t-1-t(1-\delta))\mu} = \left(\frac{e^{-\delta}}{(1-\delta)^{(1-\delta)}}\right)^{\mu}$

(choose $t = \ln(1-\delta)$)

Chernoff Bound



- Chernoff bound: Let $X_1, \ldots, X_n \in \{0,1\}$ be independent trials

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(Poisson binomial RV with mean μ)

- For any $\delta > 0$,

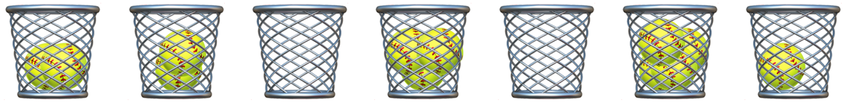
$$\Pr(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{(1+\delta)}} \right)^{\mu} \leq \begin{cases} e^{-\frac{\mu\delta^2}{3}} & \text{if } 0 < \delta < 1 \\ 2^{-(1+\delta)\mu} & \text{if } (1+\delta) \geq 2e \end{cases}$$

- For any $0 < \delta < 1$,

$$\Pr(X \leq (1-\delta)\mu) \leq \left(\frac{e^{-\delta}}{(1-\delta)^{(1-\delta)}} \right)^{\mu} \leq e^{-\frac{\mu\delta^2}{2}}$$

Balls into Bins

(Multinomial distribution)



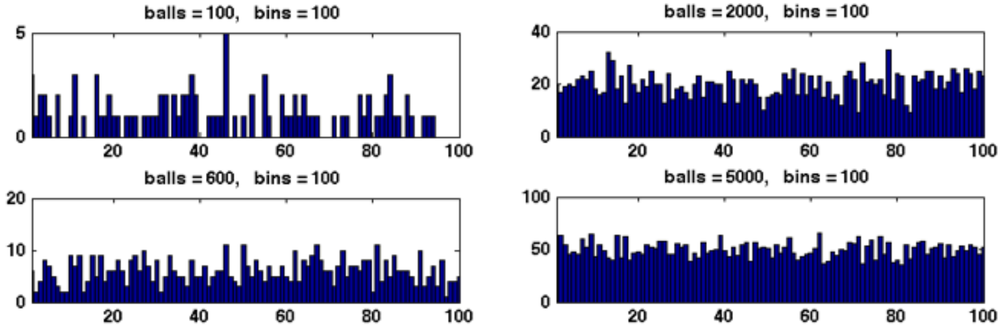
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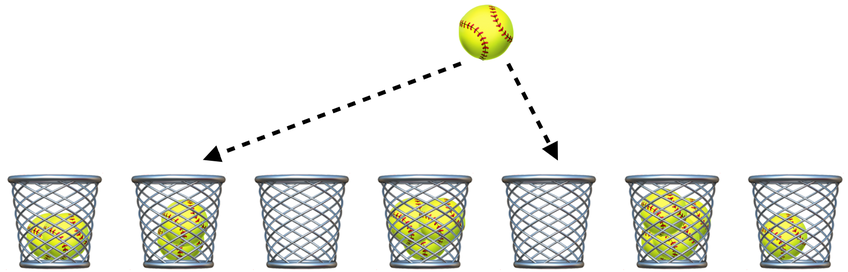
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$$\max_{1 \le i \le n} X_i = \begin{cases} O\left(\dfrac{\log n}{\log \log n}\right) & \text{if } m = n \\ O\left(\dfrac{m}{n}\right) & \text{if } m \ge n \ln n \end{cases}$$

w.h.p. (with prob. $1 - o\left(\frac{1}{n}\right)$)



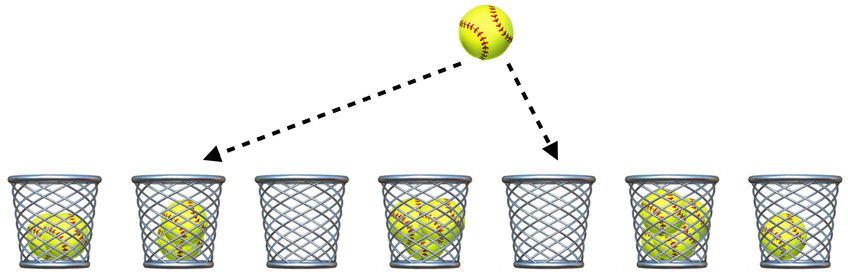
Power of two choices



- Can we do better if we are equipped with **two** random functions?
- Idea: place the ball into the bin with less load.
- Intuitively better. Proof?
- Intuition: $B_i \triangleq \sum_x I(\text{Load}(x) \geq i)$
  - $\mathbb{E}[B_i] \leq \sum_x \Pr[\text{Load}(x) \geq i] = n \cdot \Pr[\text{Load}(1) \geq i]$

$$\leq n \cdot (B_{i-1}/n)^2 = B_{i-1}^2/n$$

Power of two choices



- Can we do better if we are equipped with **two** random functions?
- Idea: place the ball into the bin with less load.
- Intuition: $B_i \triangleq \#\{\mathrm{Load}(x) \geq i\} \approx n \cdot (B_{i-1}/n)^2 = B_{i-1}^2/n$
- $B_i/n \approx (B_{i-1}/n)^2$ and $B_2/n \leq 1/2$
- Intuitively: $\dfrac{B_{2+j}}{n} \leq \dfrac{1}{2^{2^j}}$. Max-load m satisfies $B_m/n < 1$.
- $2^{2^{\log\log n}} = n \implies m \approx \log\log n$

Power of two choices

Induction-like proof strategy: layer by layer

- Let $\beta_6 = n/2e$, $\beta_{i+1} = 2e\beta_i^2/n$ be thresholds, ν_i the #bins with load $\geq i$
- Let $\mathscr{E}_i$ be event $\nu_i \leq \beta_i$. **Goal**: $\Pr[\wedge_i \mathscr{E}_i] = \Pi_i \Pr[\mathscr{E}_i \mid \mathscr{E}_{<i}] \geq 1 - O(1/n)$

- **Observation**: Given $\mathscr{E}_i$, $\nu_{i+1} \leq$ #{balls with height $> i$} $\lessapprox \mathrm{Bin}(n, \beta_i^2/n^2)$
- $\Pr[\mathrm{Bin}(n, \beta_i^2/n^2) \geq \beta_{i+1}] \leq 2^{-2e\beta_i^2/n}$ Chernoff: $\Pr(X \geq L) \leq 2^{-L}$ if $L \geq 2e\mu$
  - $\beta_i^2/n \geq \log_2 n \implies \Pr[\neg \mathscr{E}_{i+1} \mid \mathscr{E}_i] \leq 1/n^{2e}$;
  - o.w. $\nu_i < \sqrt{n \log_2 n}$, ν_{i+1} shrinks drastically: $\Pr[\nu_{i+1} \geq 6 \log n \mid \mathscr{E}_i] \leq 1/n^2$

$$\Pr[\nu_{i+2} \geq 1 \mid \nu_{i+1} < 6 \log n] \leq \Pr[\mathrm{Bin}(n, 6 \log n/n)^2 \geq 1] = O(\log^2 n)/n$$

Hashing with Chaining

- `Search(k)` where k is a key. ---------- Time depends on length of the linked list!

  Search can cost $\Theta(n)$ in worst-case. (All keys hash to same value.)

  - Compute $h(k)$
  - Go through the corresponding list to search item with key k.

- `Insert(x)` where x is a pointer to an item. ---------- $O(1)$ Note: we assume computing h takes $O(1)$ time.
  - Compute $h(x.key)$
  - Insert x to the head of the corresponding list.

- `Remove(x)` where x is a pointer to an item. ---------- $O(1)$
  - Simply remove x from the linked list.

Birthday Paradox

Paradox:
(i) a statement that leads to a contradiction;
(ii) a situation which defies intuition.



In a class of m>57 students, with >99% probability, there are two students with the same birthday.

Assumption: birthdays are uniformly & independently distributed.

n balls are thrown into m bins:
event $\mathscr{E}$: each bin receives ≤ 1 balls

Birthday Paradox

n balls are thrown into m bins:
event $\mathscr{E}$: each bin receives ≤ 1 balls

$$\Pr[\mathscr{E}] = \frac{\left|[n] \xrightarrow{1-1} [m]\right|}{\left|[n] \to [m]\right|} = \frac{m(m-1)\cdots(m-n+1)}{m^n}$$

$$= \prod_{i=0}^{n-1} \left(1 - \frac{i}{m}\right)$$

Birthday Paradox

n balls are thrown into m bins:
event $\mathscr{E}$: each bin receives ≤ 1 balls

Suppose that balls are thrown one-by-one:

$$\Pr[\mathscr{E}] = \Pr[\text{all } n \text{ balls are thrown into ditinct bins}]$$

chain rule

$$= \prod_{i=1}^{n} \Pr[\text{the } i\text{th ball is thrown into an empty bin} \mid \text{first } i-1 \text{ balls are thrown into ditinct bins}]$$

$$= \prod_{i=1}^{n} \left(1 - \frac{i-1}{m}\right) = \prod_{i=0}^{n-1} \left(1 - \frac{i}{m}\right)$$

Birthday Paradox

n balls are thrown into m bins:
event $\mathscr{E}$: each bin receives ≤ 1 balls

(Taylor: $1 - x \approx \mathrm{e}^{-x}$ for $x = o(1)$)

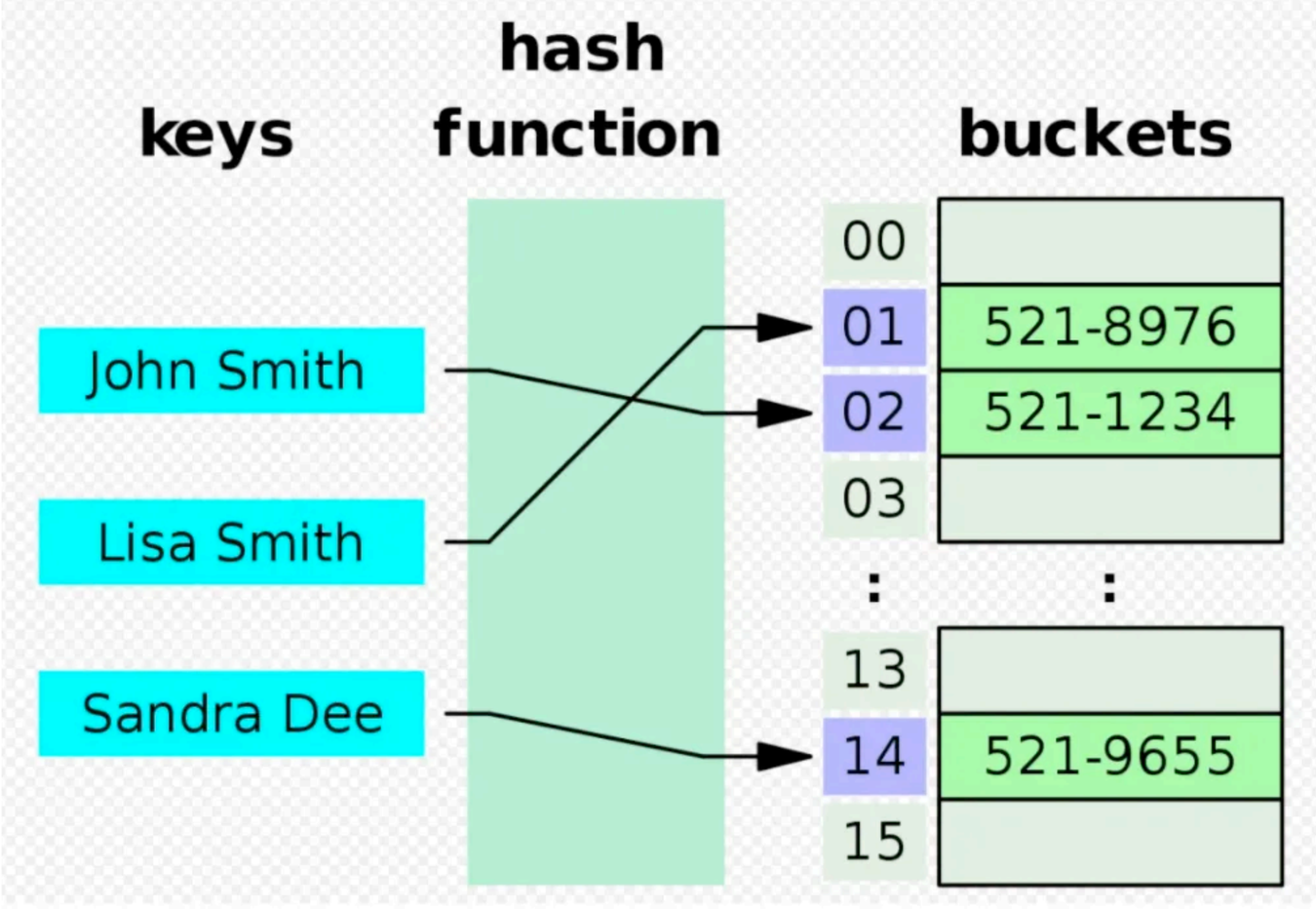
$$\Pr[\mathscr{E}] = \prod_{i=0}^{n-1}\left(1 - \frac{i}{m}\right) \approx \prod_{i=0}^{n-1} \mathrm{e}^{-\frac{i}{m}} \approx \mathrm{e}^{-n^2/2m}$$

Formally: $\mathrm{e}^{-(1+o(1))n^2/2m} \leq \prod_{i=0}^{n-1}\left(1 - \frac{i}{m}\right) \leq \mathrm{e}^{-(1-o(1))n^2/2m}$

(assuming $n \ll m$)

when $n = \sqrt{2m \ln \frac{1}{p}} \implies \Pr[\mathscr{E}] = (1 \pm o(1))p$

Hash Table


keys
hash function
buckets
John Smith
Lisa Smith
Sandra Dee
00
01
02
03
13
14
15
521-8976
521-1234
521-9655

Data Structure for Set

Data: a set S of n items $x_1, x_2, \ldots, x_n \in U = [N]$

Query: an item $x \in U$

Determine whether $x \in S$.

- **Space cost**: size of data structure (in bits)
  - entropy of a set: $\log \binom{N}{n} = O(n \log N)$ bits (when $N \gg n$)
- **Time cost**: time to answer a query (in memory accesses)
- Balanced tree: $O(n \log N)$ bits of space, $O(\log n)$ time
- Perfect hashing: $O(n \log N)$ bits of space, $O(1)$ time

Perfect Hashing

$S = \{a, b, c, d, e, f\} \subseteq [N]$ of size n

uniform random

h $[N] \to [m]$

no collision

$\Pr[\textit{perfect}] \approx e^{-n^2/2m} > 1/2$

Table T:

e	b		d		f		c	a	

$m = n^2$

Birthday

SUHA: Simple Uniform Hash Assumption

Query(x):

retrieve hash function h;

check whether $T[h(x)] = x$;

Universal Hashing

Universal Hash Family (Carter and Wegman 1979):

A family $\mathscr{H}$ of hash functions in $U \to [m]$ is **k-universal** if for any $\leq k$ distinct $x_1, \ldots, x_k \in U$,

$$\Pr_{h \in \mathscr{H}} \left[h(x_1) = \cdots = h(x_k) \right] \leq \frac{1}{m^{k-1}}.$$

Moreover, $\mathscr{H}$ is **strongly k-universal** (k-wise independent) if for any $\leq k$ distinct $x_1, \ldots, x_k \in U$ and any $y_1, \ldots, y_k \in [m]$,

$$\Pr_{h \in \mathscr{H}} \left[\bigwedge_{i=1}^{k} h(x_i) = y_i \right] = \frac{1}{m^k}.$$

k-Universal Hash Family

hash functions $h : U \to [m]$

- Linear congruential hashing:
  - Represent $U \subseteq \mathbb{Z}_p$ for sufficiently large prime p
  - $h_{a,b}(x) = ((ax + b) \bmod p) \bmod m$
  - $\mathscr{H} = \left\{ h_{a,b} \mid a \in \mathbb{Z}_p \backslash \{0\}, b \in \mathbb{Z}_p \right\}$

> **Theorem**:
> The linear congruential family $\mathscr{H}$ is 2-wise independent.

- Degree-k polynomial in finite field with random coefficients
- Hashing between binary fields: $GF(2^w) \to GF(2^l)$

$$h_{a,b}(x) = \texttt{(a*x+b)>>(w-l)}$$

Birthday Paradox (pairwise independence)

n balls are thrown into m bins: by 2-universal hashing

event $\mathscr{E}$: each bin receives ≤ 1 balls

- Location of n balls: $X_1, X_2, \ldots, X_n \in [m]$
- Total # of collisions:

$$Y = \sum_{i<j} I[X_i = X_j]$$

- **Linearity of expectation:**

$$\mathbb{E}[Y] = \sum_{i<j} \Pr[X_i = X_j] \quad \leq \binom{n}{2} \frac{1}{m}$$

2-universal

Birthday Paradox (pairwise independence)

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2-universal

when $n \leq \sqrt{2m\epsilon}$

- **Markov's inequality:** $\Pr[\neg \mathscr{E}] = \Pr[Y \geq 1] \leq \mathbb{E}[Y] \leq \epsilon$

Perfect Hashing

$S = \{a, b, c, d, e, f\} \subseteq [N]$ of size n

2-universal h $[N] \to [m]$ Pr[*imperfect*] $= \dfrac{n(n-1)}{2m}$

Table T:

e	b		d		f		c	a	

m

For 2-universal family $\mathscr{H}$ from $[N]$ to $[m]$, if $m > \binom{n}{2}$, for any $S \subseteq [N]$ of size n, there is an $h \in \mathscr{H}$ that cause no collisions over S.

Query(x):

retrieve hash function h;

check whether $T[h(x)] = x$;

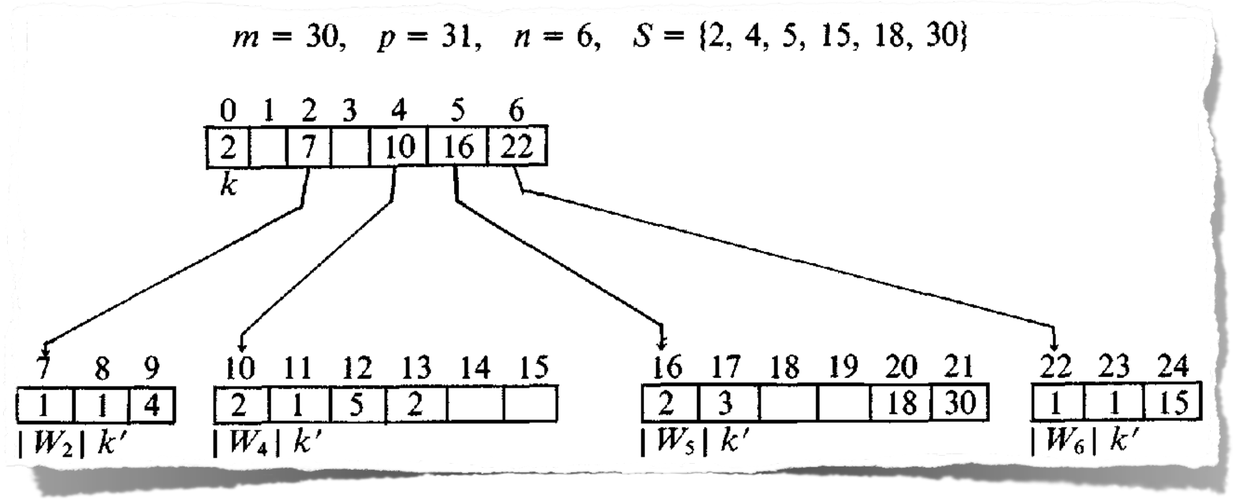
FKS Perfect Hashing

(Fredman, Komlós, Szemerédi, 1984)

Data: a set S of n items $x_1, x_2, \ldots, x_n \in U = [N]$

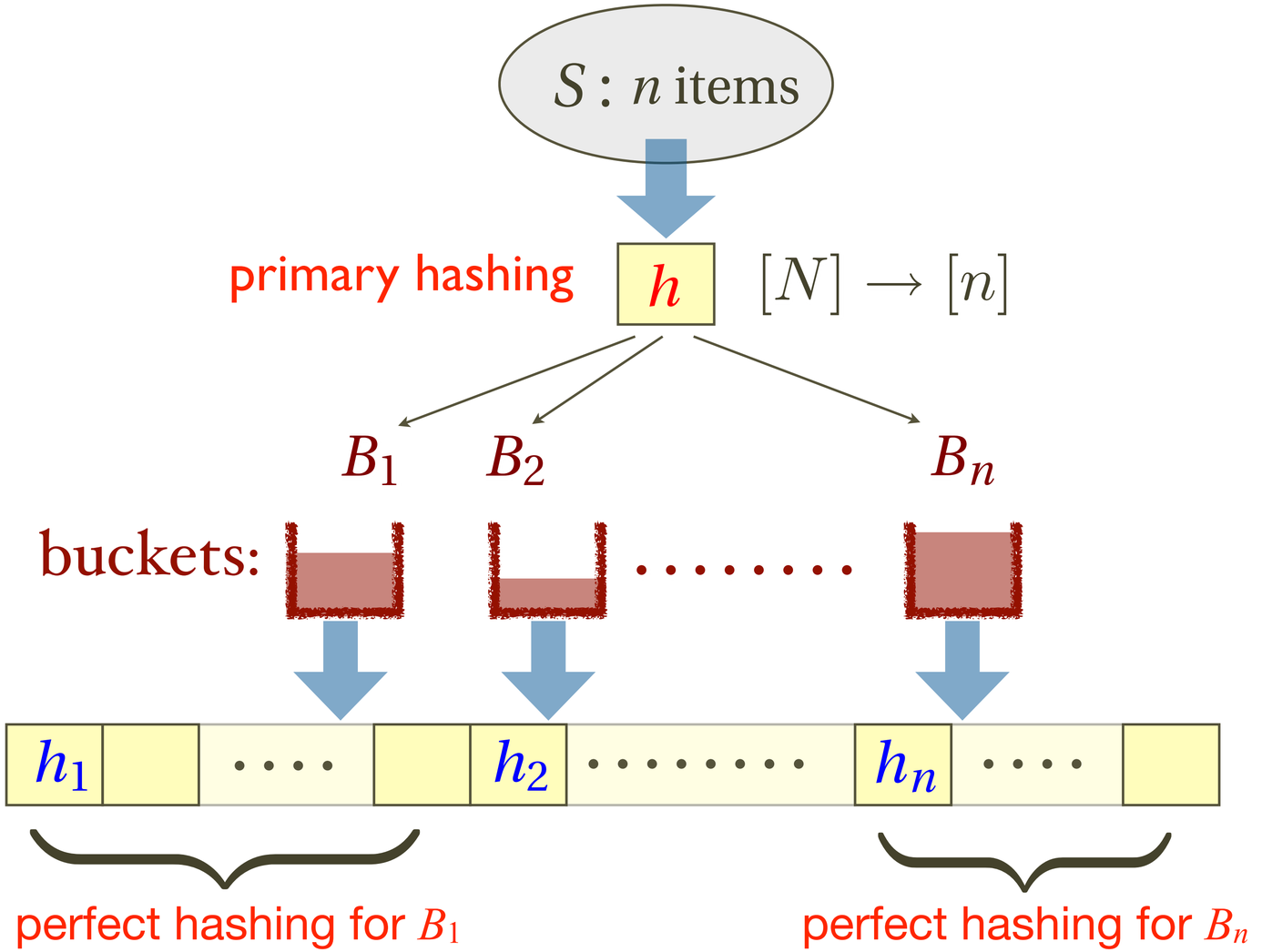
Query: an item $x \in U$

Determine whether $x \in S$.



- **Space cost**: $O(n)$ words (each of $O(\log N)$ bits)
- **Time cost**: $O(1)$ for each query in the worst case

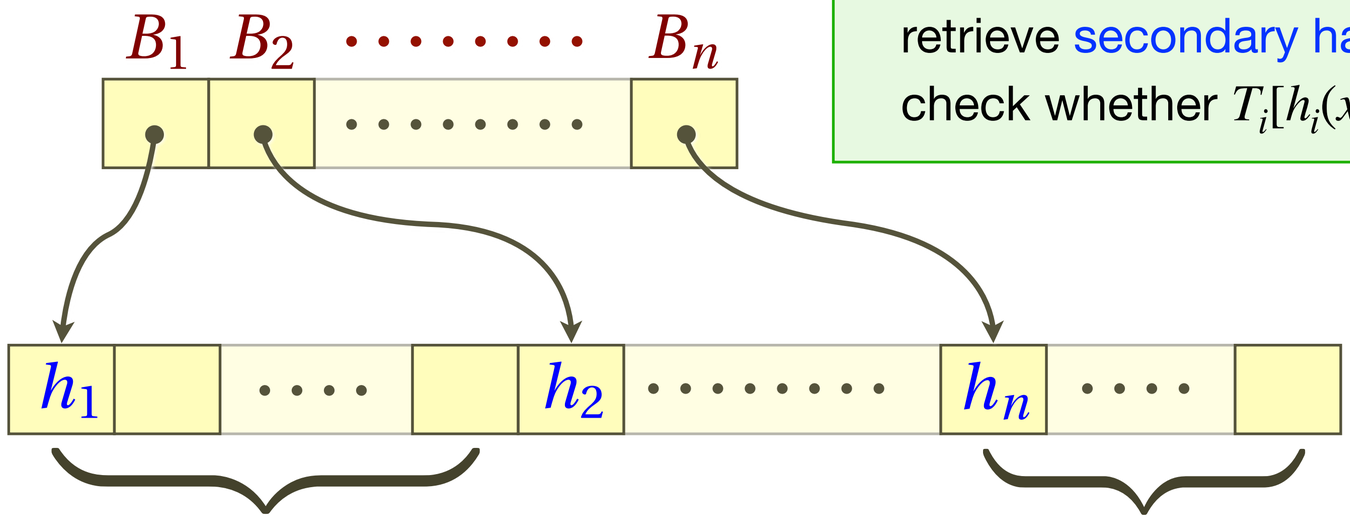
FKS Perfect Hashing



FKS Perfect Hashing

Set $S \subseteq [N]$ of size n





perfect hashing for B_1 using space $|B_1|^2$

perfect hashing for B_n using space $|B_n|^2$

Query(x):

retrieve primary hash h;

goto bucket $i = h(x)$;

retrieve secondary hash h_i;

check whether $T_i[h_i(x)] = x$;

- $\exists\ h_1, \ldots, h_n$ from 2-universal family s.t. h_i is perfect for B_i for all i

Collision Number

n balls are thrown into m bins by 2-universal hashing

- Location of n bins: $X_1, X_2, \ldots, X_n \in [m]$

Collision #: $Y = \sum_{i<j} I[X_i = X_j]$

- Linearity of expectation:

$$\mathbb{E}[Y] = \sum_{i<j} \Pr[X_i = X_j] \quad \leq \binom{n}{2} \frac{1}{m}$$

2-universal

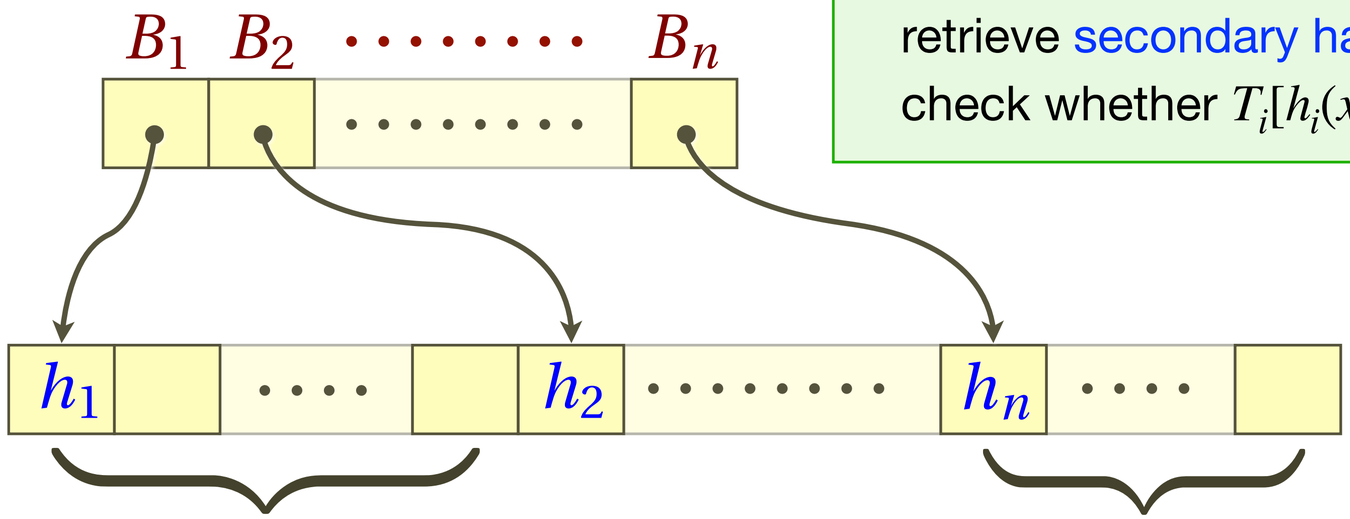
- Size of the i-th bin: $|B_i|$

$$Y = \sum_{i=1}^{n} \binom{|B_i|}{2} = \frac{1}{2} \sum_{i=1}^{n} |B_i|(|B_i| - 1) \Longrightarrow \mathbb{E}\left[\sum_{i=1}^{n} |B_i|^2\right] = \frac{n(n-1)}{m} + n$$

FKS Perfect Hashing

Set $S \subseteq [N]$ of size n





perfect hashing for B_1 using space $|B_1|^2$

perfect hashing for B_n using space $|B_n|^2$

Query(x):

retrieve primary hash h;

goto bucket $i = h(x)$;

retrieve secondary hash h_i;

check whether $T_i[h_i(x)] = x$;

- $\exists h$ from a 2-universal family s.t. the total space cost is O(n)

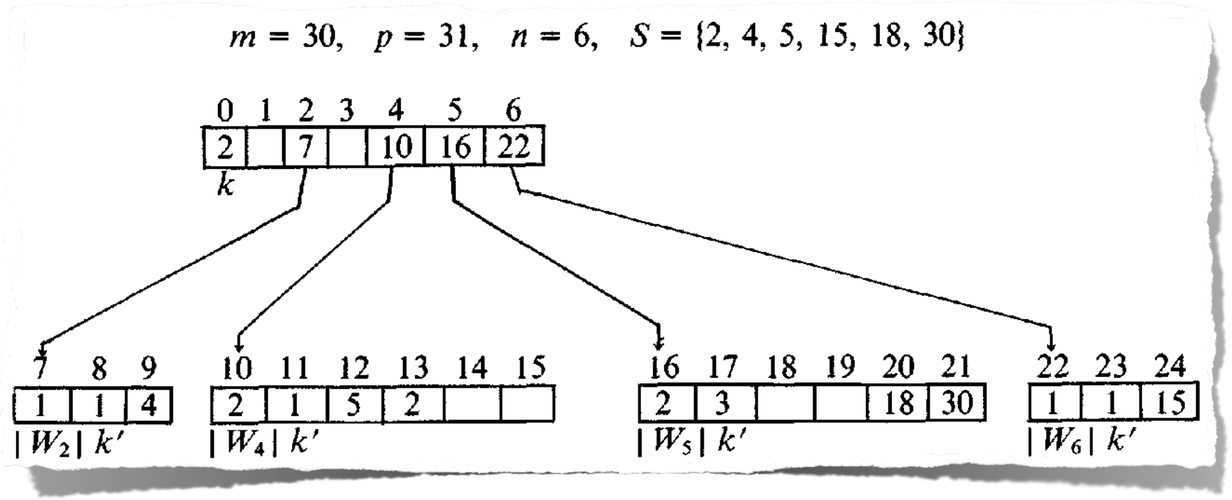
FKS Perfect Hashing

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Data: a set S of n items $x_1, x_2, \ldots, x_n \in U = [N]$

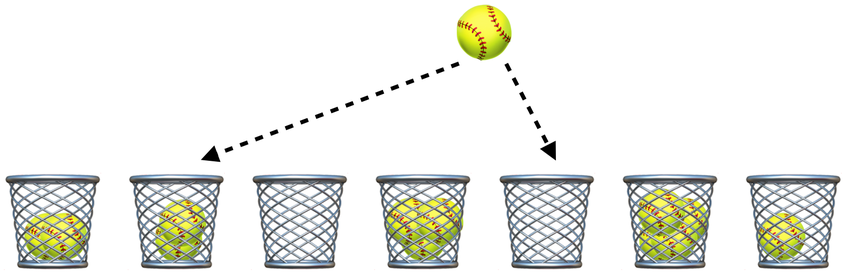
Query: an item $x \in U$

Determine whether $x \in S$.



- $O(n \log N)$ space, $O(1)$ time in the worst case
- Dynamic version: **[Dietzfelbinger, Karlin, Mehlhorn, Meyer auf der Heide, Rohnert, Tarjan, 1984]**

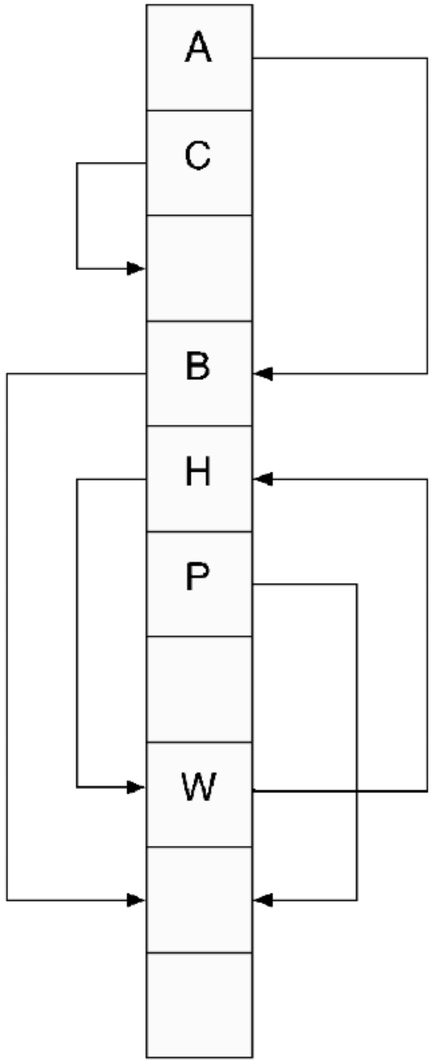
Chaining with two choices



- Maximum load by the power of two choices: $O(\log \log n)$
- $O(\log \log n)$ time for chaining: enumerating because of collisions & max load
- A hash table with two functions?
  - Item x must be placed in $h_1(x)$ or $h_2(x)$, time cost: exact 2
- Collision? Both of two bins are occupied?
  - Kick them away!



Cuckoo Hashing



Cuckoo Hashing:

Hash table $A[1 \ldots m]$ initialized to empty

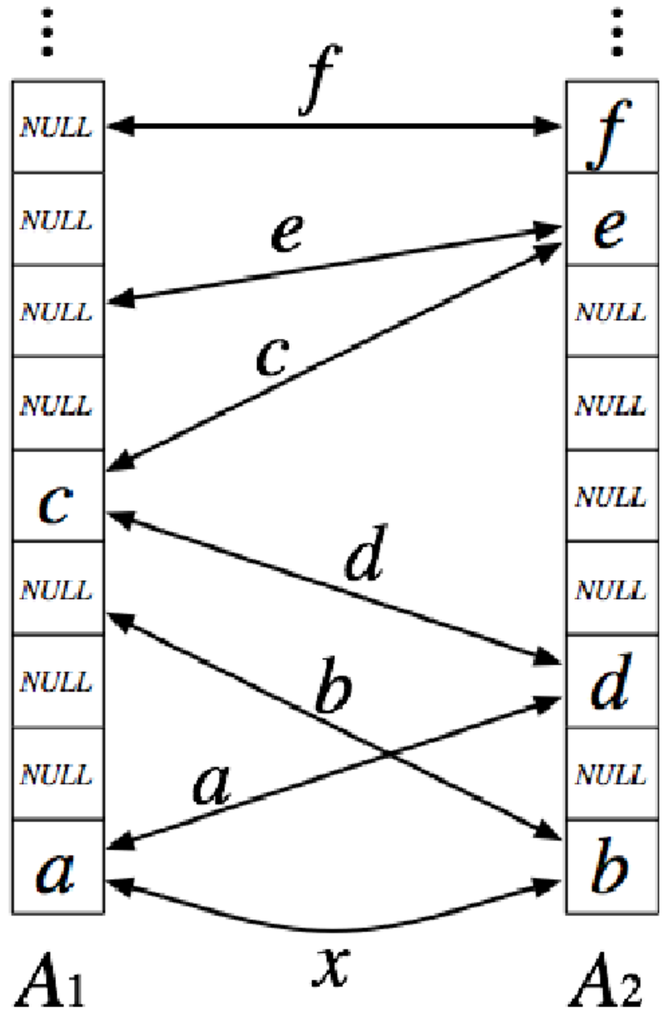
Query x: examine $A[h(x)], A[g(x)]$

Insert x: place x if exists spare slot;

o.w. while no spare slot, keep kicking;

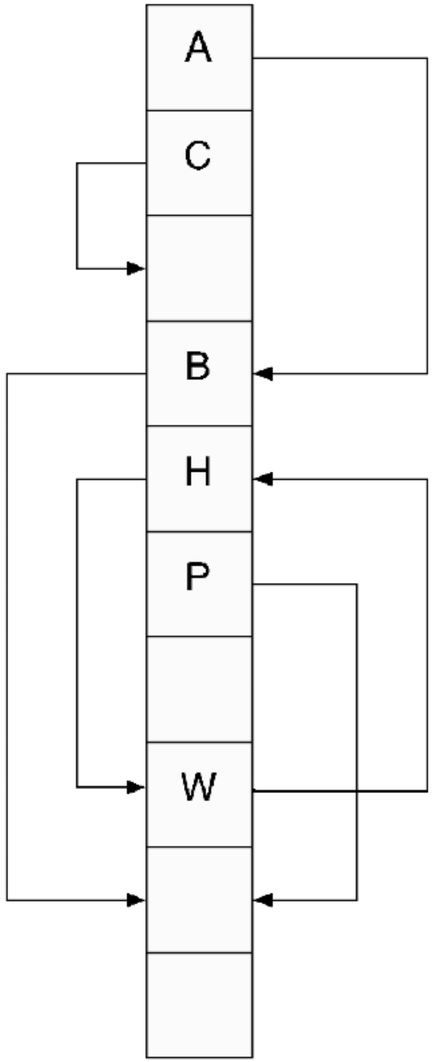
rebuild if take so long

- Space cost: $O(n)$
- Query time: 2
- Update time: ?



Cuckoo Graph

Analysis of Cuckoo hashing



Cuckoo Hashing:

Hash table $A[1 \ldots m]$ initialized to empty

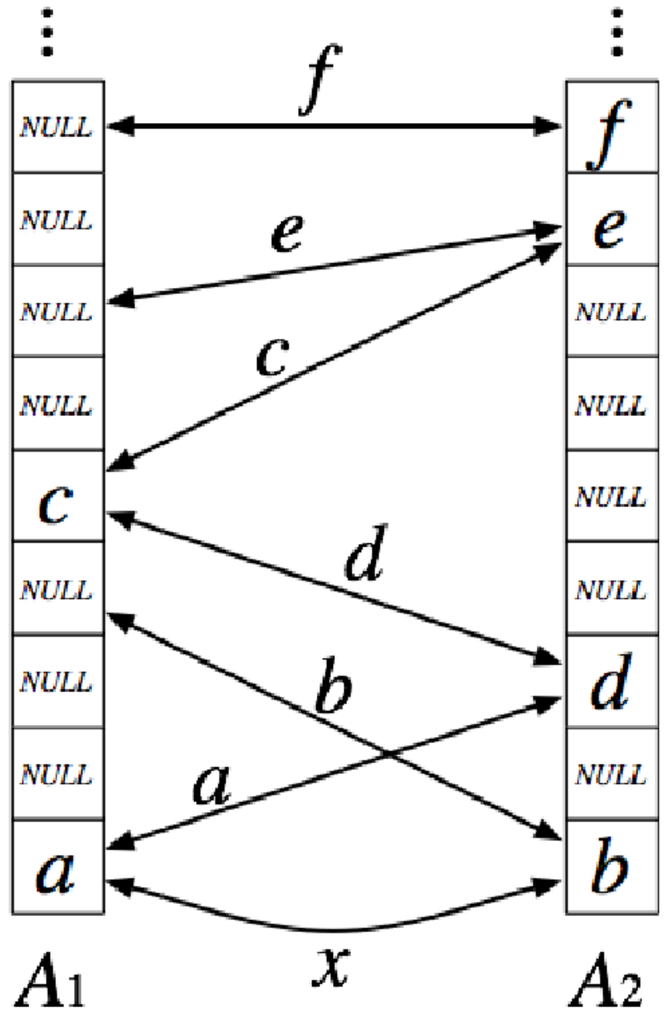
Query x: examine $A[h(x)], A[g(x)]$

Insert x: place x if exists spare slot;

o.w. while no spare slot, keep kicking;

rebuild if take so long

- A multigraph $G = (V, E)$
  - $V =$ all the slots
  - $E = \{(h(x), g(x)) : x \in S\}$
    - Edge labeled by insertion



Cuckoo Graph

Analysis of Cuckoo hashing

- A multigraph $G = (V, E)$
  - $V =$ all the slots
  - $E = \{(h(x), g(x)) : x \in S\}$

- When does insertion succeed?
  - We have a kicking path:
  - Or, a single kicking cycle:

- When does insertion fail?
  - A double kicking cycle:
  - Long path

Cuckoo Hashing:

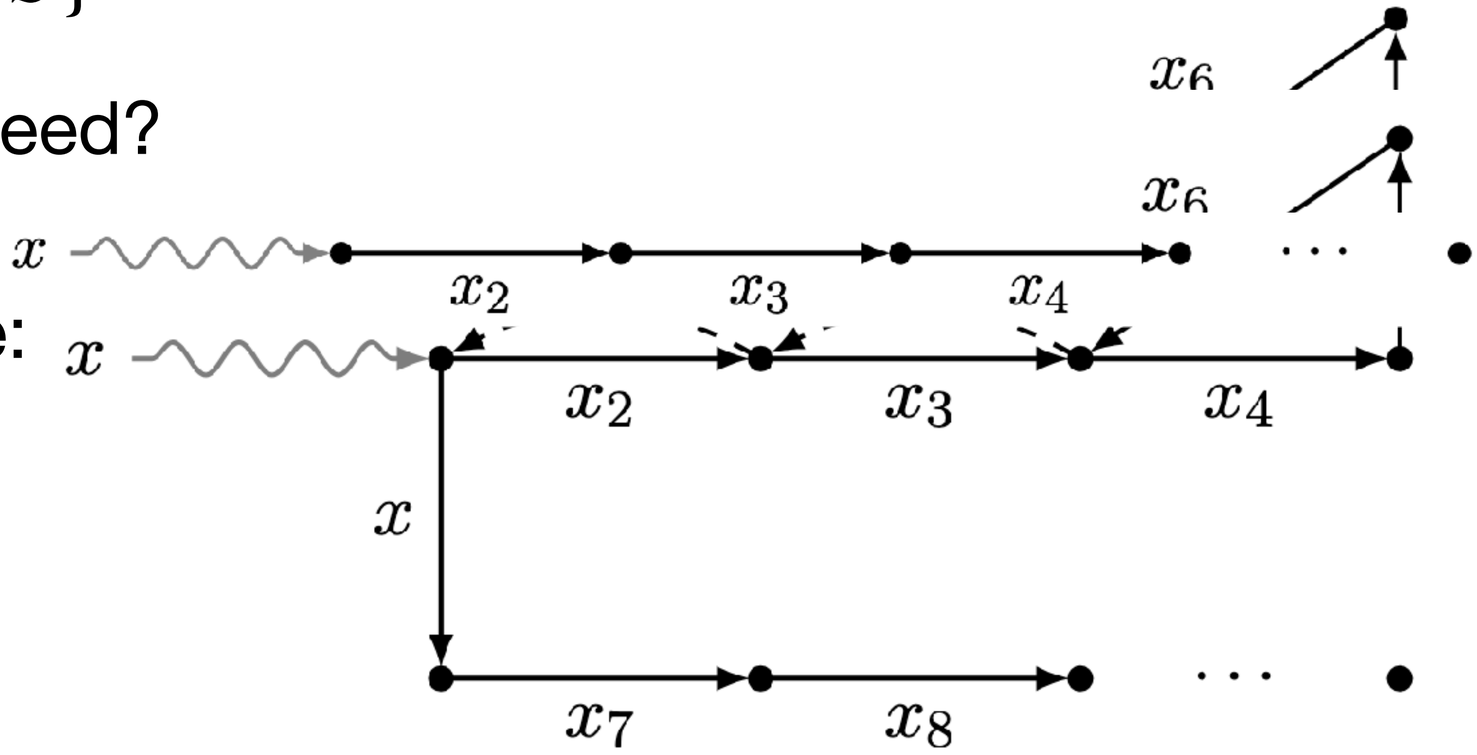
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Query x: examine $A[h(x)], A[g(x)]$

Insert x: place x if exists spare slot;

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rebuild if take so long



Cuckoo Graph

Analysis of Cuckoo hashing

- path 👍
- single cycle 👍
- double cycle 👎

Cuckoo Hashing:

Hash table $A[1 \ldots m]$ initialized to empty

Query x: examine $A[h(x)], A[g(x)]$

Insert x: place x if exists spare slot;

o.w. while no spare slot, keep kicking;

rebuild if take so long

- $\mathbb{E}[T] \leq \mathbb{E}[\text{len(path)}] + \mathbb{E}[\text{len(cycle)}] + \Pr[\text{d-cycle, long path}] \cdot n \cdot \mathbb{E}[T]$
- Claim: A fixed insertion path of length k exists w.p. $\leq m^{-2k-1}$
  - Path be like $x,1,b,2,c,3,e,4,\ldots$ key-slot-key-slot-…
- $\Pr[\text{len(path)} \geq k] \leq (nm)^k/m^{2k+1} \leq n^k/m^{k+1} = \exp(-\Omega(k))$
- $\Pr[\text{len(cycle)} \geq k] = \exp(-\Omega(k))$, $\Pr[\text{path} \geq 2\log n] \leq 1/n^{\Omega(1)}$
- $\mathbb{E}[\text{len(path)}]$, $\mathbb{E}[\text{len(cycle)}] \leq \Sigma_{k \geq 1} \exp(-\Omega(k)) = O(1)$

Cuckoo Graph

Analysis of Cuckoo hashing

- path 👍
- single cycle 👍
- double cycle 👎

Cuckoo Hashing:

Hash table $A[1\ldots m]$ initialized to empty

Query x: examine $A[h(x)], A[g(x)]$

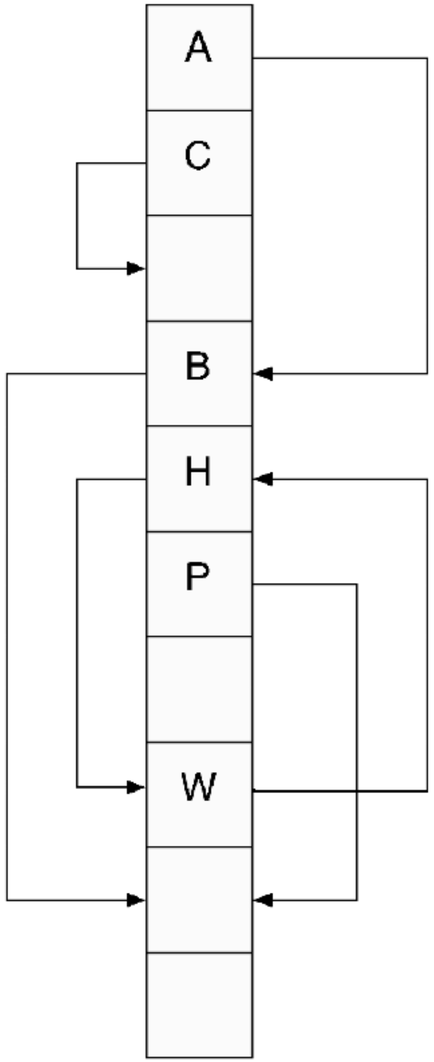
Insert x: place x if exists spare slot;

o.w. while no spare slot, keep kicking;

rebuild if take so long

- $\mathbb{E}[T] \le O(1) + O(1) + (\Pr[\text{d-cycle}] + 1/n^2) \cdot n \cdot \mathbb{E}[T]$
- Claim: A fixed insertion path of length k exists w.p. $\le m^{-2k-1}$
  - Path be like $x,1,b,2,c,3,e,4,\ldots$ key-slot-key-slot-…
- $\Pr[\text{d-cycle}]$?
  - Observe. Simplest case ⦶: $\Pr \le n^2m^2/m^6$.
  - General case ⊶: #edge ≤ #slot-1. $\Pr \le n^k m^k/m^{2k+2}$

Cuckoo Hashing



Cuckoo Hashing:

Hash table $A[1 \ldots m]$ initialized to empty

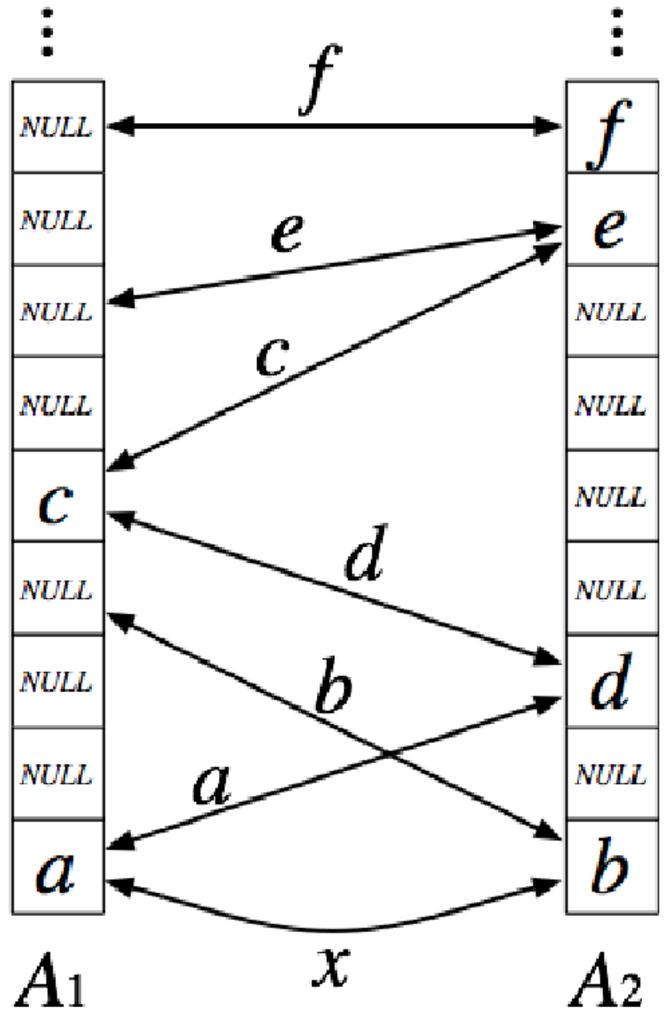
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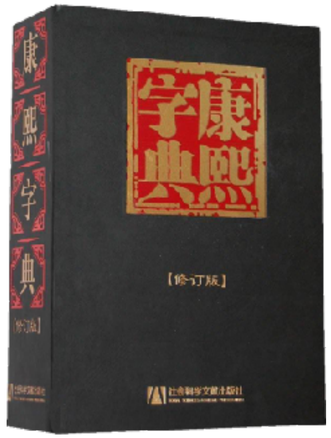
rebuild if take so long

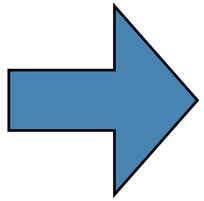
- Space cost: $O(n)$
- Query time: 2
- Update time: expected amortized $O(1)$



Static Dictionary

Goal: O(1) search time







(word,meaning)-pair

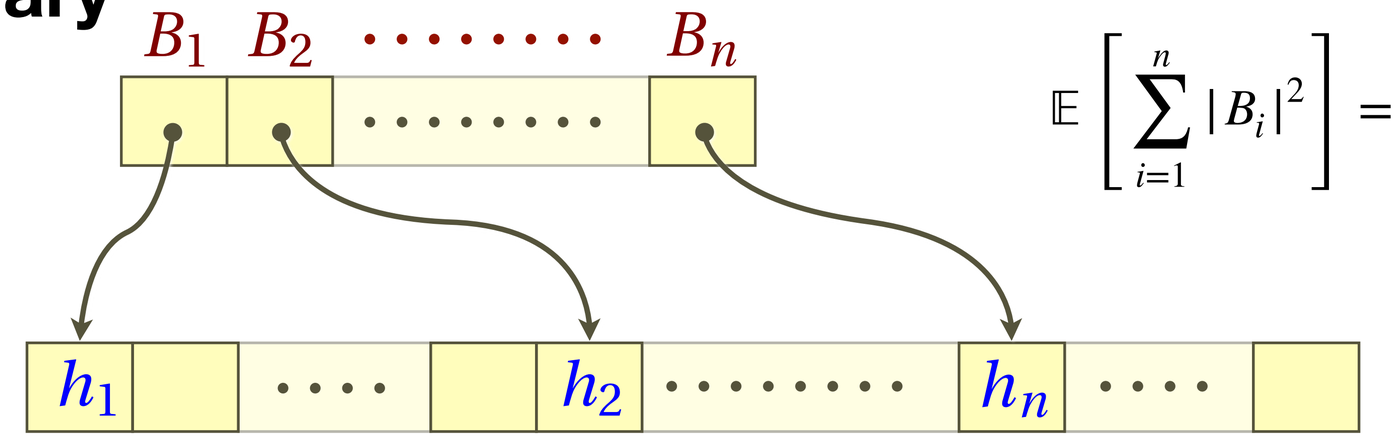
(key,value)-pair

Succinct Dictionaries

	Space	Query time	Update time	
Binary search tree	$O(n)$	$O(\log n)$	$O(\log n)$	bad time
Hashing with Chaining	$O(n)$	expected $O(1)$	expected $O(1)$	bad time
1-level Perfect hashing	$O(n^2)$	1	1	Huge space
2-level Perfect hashing	$O(n)$	$O(1)$	$O(1)$	Huge constants
Cuckoo hashing	$O(n)$	2	expected amortized $O(1)$	worst case?
Succinct Dictionary	$(1+o(1))n$	$O(1)$	$O(1)$	Best of both worlds

Succinct Dictionaries

Small Dictionary



$$\mathbb{E}\left[\sum_{i=1}^{n} |B_i|^2\right] = \frac{n(n-1)}{m} + n$$

- Exactly, how much space do we need for FKS hashing?
  - First-level pointers & hashing ($n \log n$ bits) + space for raw data (nv bits)
- Simple observation: it is succinct if $\log n \ll v$
  - Say, $v = \log_2 U$. It is succinct if we use it to maintain $n = \text{poly} \log U$ keys

Succinct Dictionaries

First attempt

- Examine the balls-into-bins argument:
  - n balls into $n/\log^3 n$ bins. Let $n' \triangleq (1 + 2/\log n)\log^3 n$
  - Chernoff bound: $\Pr[\text{load} \geq n'] \leq \exp(-\mu\delta^2/2) = n^{-2}$
  - Maximum load $\leq (1 + 2/\log n)\log^3 n$ w.h.p.
- Each bin is a small dictionary with redundancy $n' \log n'$ bits
- Overall redundancy $n \log n' + (2/\log n) n \log U = o(n \log U)$ bits

any issue?

Succinct Dictionaries

Better small dictionary 😀 & Let it overflow!

- Intuitively, we can maintain $O(\log n/\log\log n)$ keys with $O(\log n)$ bits
  - e.g. binary search tree
  - Intuitively, any operation can be done in $O(1)$ times as the word size is $\Omega(\log n)$ bits.
- balls-into-bins argument:
  - n balls into n/m bins, where $m = \log n/\log\log n$
  - Chernoff bound: $\Pr[\text{load} \geq (1+\delta)m] \leq \exp(-m\delta^2/2) = \log^{-\Omega(1)} n$
  - #overflowing $\approx n/\text{poly}\log n$

Let $\delta = \Omega(\log\log n/\sqrt{\log n})$

A dictionary maintains $n/\log^9 n$ keys with $n/\log n$ bits

Succinct Dictionaries

Trie

- We can maintain $O(\log n / \log \log n)$ keys with $O(\log n)$ bits
  - Most basic operations can be done in $O(1)$ times with an $n^{1/3}$ bits precomputed lookup table, since the word size is $\Omega(\log n)$ bits.
- Trie: Elias-Fano in fact
  - Compressed trie: upon insertion, reaches leaf, grows until splits
  - Assume random keys
    - One insertion grows k times w.p. 2^{-k}, geometrically.
    - Insertion n' keys, #grows $\sim$ negative binomial.
    - #grows $= \Omega(\log n)$ w.p. $\exp(-\Omega(\log n)) = n^{-\Omega(1)}$

A binary tree of $O(\log n)$ nodes!

encode with $O(\log n)$ bits!

Succinct Dictionaries

Dealing the overflowing

- A dictionary maintains $n/\log^9 n$ keys with $n/\log n$ bits
  - n' balls into n' bins, where $n' = n/\log^9 n$.
  - $\Pr[\text{load} \geq m] \leq \sum_{k \geq m} \binom{n'}{k} \left(\frac{1}{n'}\right)^k = \exp(-\Omega(m \log n'))$
  - Let $m = \log n / \log \log n$, the probability is $\exp(-\Omega(\log n))$
  - Apply the trie
- Overall space: $O(n' \log n) = O(n/\log n)$ bits

Succinct Dictionaries

Better small dictionary 😀 & Let it overflow!

- We can maintain $O(\log n/\log\log n)$ keys with $O(\log n)$ bits
  - Most basic operations can be done in $O(1)$ times with an $n^{1/3}$ bits precomputed lookup table, since the word size is $\Omega(\log n)$ bits.
- balls-into-bins argument:
  - n balls into n/m bins, where $m = \log n/\log\log n$
  - Chernoff bound: $\Pr[\text{load} \geq (1+\delta)m] \leq \exp(-m\delta^2/2)$
  - #overflowing $\approx n/\text{poly}\log n$
  - Redundancy: $O(\delta n)$ words

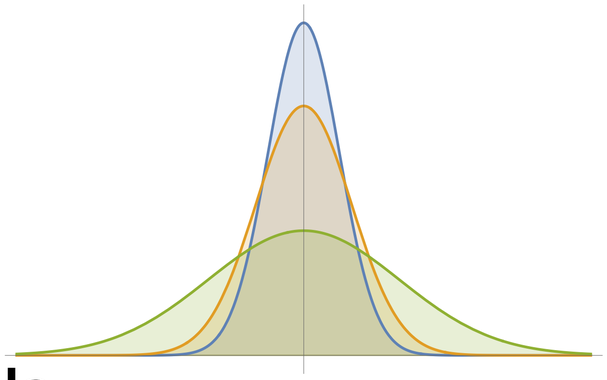
Let $\delta = \Omega(\log\log n/\sqrt{\log n})$

A dictionary maintains $n/\log^9 n$ keys with $n/\log n$ bits

$O(\delta n \log(u/n))$ bits by Feistel

Hashing in Practice

Chernoff Bound



- Chernoff bound: Let $X_1, \ldots, X_n \in \{0,1\}$ be **independent** trials

$$X = \sum_{i=1}^{n} X_i \quad \text{and} \quad \mu = \mathbb{E}[X]$$

(Poisson binomial RV with mean μ)

- For any $\delta > 0$,

$$\Pr(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{(1+\delta)}} \right)^{\mu}$$

- For any $0 < \delta < 1$,

$$\Pr(X \leq (1-\delta)\mu) \leq \left(\frac{e^{-\delta}}{(1-\delta)^{(1-\delta)}} \right)^{\mu}$$

Chernoff Bound With Limited Independency

Save randomness, save money

- **Chernoff bound**: Let $X_1, \ldots, X_n \in \{0,1\}$ be ***k*-wise independent** trials

$$X = \sum_{i=1}^{n} X_i \quad \text{and} \quad \mu = \mathbb{E}[X]$$

- For any $\delta > 0$, $k \leq \dfrac{\mu\delta}{1 - \mu/n}$,

$$\Pr(X \geq (1+\delta)\mu) \leq \frac{\binom{n}{k}(\mu/n)^k}{\binom{(1+\delta)\mu}{k}} = \left(\frac{\mu}{n}\right)^k \cdot \prod_{i=0}^{k-1} \frac{n-i}{(1+\delta)\mu - i}$$

A closer look at Chernoff

- Why do we like **moment generating function**?
  - $M_x(t) = \mathbb{E}\left[e^{tX}\right] = \sum_{i \geq 0}^{\infty} t^i \mathbb{E}\left[X^i\right]/i!$ exhausts all **moments**

- What is k-th moment of sum of Bernoulli R.V.?
  - $X^1 = \sum_i X_i,\ X^2 = \sum_i X_i^2 + \sum_{i \neq j} X_i X_j = X^1 + \sum_{i \neq j} X_i X_j$
  - $X^3 = \sum_i X_i^3 + \sum_{i \neq j} (X_i X_j^2 + X_i^2 X_j) + \sum_{i \neq j \neq k} X_i X_j X_k = \sum_i X_i + 2\sum_{i \neq j} X_i X_j + \sum_{i \neq j \neq k} X_i X_j X_k$

- Intuition?
  - $X^k = \sum_{i \leq \min\{k,n\}} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \ldots j_i \leq n} X_{j_1} X_{j_2} \ldots X_{j_i}$

A closer look at Chernoff

- Why do we like **moment generating function**?
  - $M_x(t) = \mathbb{E}\left[e^{tX}\right] = \sum_{i \geq 0}^{\infty} t^i \mathbb{E}\left[X^i\right] / i!$ exhausts all **moments**
- Intuition?
  - $e^{tX} = \sum_{i \leq \min\{k,n\}} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \ldots j_i \leq n} X_{j_1} X_{j_2} \ldots X_{j_i}$ for some $\alpha_1, \ldots, \alpha_n \geq 0$
  - $\Pr[X \geq a] = \Pr\left[\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \ldots j_i \leq n} X_{j_1} X_{j_2} \ldots X_{j_i} \geq \sum_{i=0}^{a} \alpha_i \binom{a}{i}\right]$

Generalized Chernoff

- $\Pr[X \geq a] = \Pr\left[\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \ldots j_i \leq n} X_{j_1} X_{j_2} \ldots X_{j_i} \geq \sum_{i=0}^{a} \alpha_i \binom{a}{i}\right]$
- For Chernoff, the α_i's are fixed s.t. $\sum_{i \leq n} \alpha_i \sum_{1 \leq j_1 < j_2 < \ldots j_i \leq n} X_{j_1} X_{j_2} \ldots X_{j_i} = \sum_{i \leq n} \frac{t^i X^i}{i!}$
- Can we do better from this perspective?
  - Parameterize α_i's instead of simply t

Generalized Chernoff

- $\Pr[X \geq a] = \Pr\left[\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \ldots j_i \leq n} X_{j_1} X_{j_2} \ldots X_{j_i} \geq \sum_{i=0}^{a} \alpha_i \binom{a}{i}\right]$
  - Parameterize α_i's instead of simply t
- Markov first: $\Pr[X \geq a] \leq \dfrac{\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \ldots j_i \leq n} \mathbb{E}\left[X_{j_1} X_{j_2} \ldots X_{j_i}\right]}{\sum_{i=0}^{a} \alpha_i \binom{a}{i}}$
- Observation: for $i > a$, α_i should be 0
- Observation: α_i should be 0 for i not minimizes $\sum \mathbb{E}\left[X_{j_1} X_{j_2} \ldots X_{j_i}\right] \Big/ \binom{a}{i}$

Generalized Chernoff

Parameterize α_i's instead of simply t

- $\Pr[X \geq a] \leq \dfrac{\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \ldots j_i \leq n} \mathbb{E}\left[X_{j_1} X_{j_2} \ldots X_{j_i}\right]}{\sum_{i=0}^{a} \alpha_i \binom{a}{i}}$
- Observation: $\sum \mathbb{E}\left[X_{j_1} X_{j_2} \ldots X_{j_i}\right]$ maximized when $p_1 = \ldots = p_n$

  subject to $\sum_i p_i = \mu$

  - $\sum_{1 \leq j_1 < j_2 < \ldots j_i \leq n} \mathbb{E}\left[X_{j_1} X_{j_2} \ldots X_{j_i}\right] \leq \binom{n}{i} (\mu/n)^i$ if mutual independence

Generalized Chernoff

Parameterize α_i's instead of simply t (assume independence)

- $\Pr[X \geq a] \leq \dfrac{\sum_{i \leq n} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \ldots j_i \leq n} \mathbb{E}\left[X_{j_1} X_{j_2} \ldots X_{j_i}\right]}{\sum_{i=0}^{a} \alpha_i \binom{a}{i}} \leq \dfrac{\sum_{i=0}^{a} \alpha_i \binom{n}{i} (\mu/n)^i}{\sum_{i=0}^{a} \alpha_i \binom{a}{i}}$
- Observation: α_i should be 0 for i not minimizes $\binom{n}{i}(\mu/n)^i \Big/ \binom{a}{i}$
- Observation: $\dfrac{\binom{n}{i+1}(\mu/n)^{i+1}}{\binom{a}{i+1}} \Big/ \dfrac{\binom{n}{i}(\mu/n)^i}{\binom{a}{i}} = \dfrac{n-i}{a-i}\dfrac{\mu}{n}$, minimizes when

$$i \triangleq (a-\mu)/(1-\mu/n)$$

Generalized Chernoff

Parameterize α_i's instead of simply t (assume independence)

- $\Pr[X \geq a] \leq \dfrac{\sum_{i=0}^{a} \alpha_i \binom{n}{i} (\mu/n)^i}{\sum_{i=0}^{a} \alpha_i \binom{a}{i}} \leq \binom{n}{k} (\mu/n)^k \Big/ \binom{a}{k}$ for all $k \leq i^*$
- Observation: α_i should be 0 for i not minimizes $\binom{n}{i} (\mu/n)^i \Big/ \binom{a}{i}$
- Observation: minimizes when $i^* \triangleq (a - \mu)/(1 - \mu/n)$

Generalized Chernoff

Parameterize α_i's instead of simply t (k-wise independence)

- $$\Pr[X \geq a] \leq \frac{\sum_{i=0}^{a} \alpha_i \cdot \sum_{1 \leq j_1 < j_2 < \ldots j_i \leq n} \mathbb{E}\left[X_{j_1} X_{j_2} \ldots X_{j_i}\right]}{\sum_{i=0}^{a} \alpha_i \binom{a}{i}} \leq \frac{\sum_{i=0}^{a} \alpha_i \binom{n}{i} (\mu/n)^i}{\sum_{i=0}^{a} \alpha_i \binom{a}{i}}$$

$$\leq \binom{n}{k} (\mu/n)^k \Big/ \binom{a}{k} \text{ for all } k \leq i^*$$

- Observation: α_i should be 0 for $i \neq i^* \triangleq (a - \mu)/(1 - \mu/n)$
- Observation: $\sum \mathbb{E}\left[X_{j_1} X_{j_2} \ldots X_{j_k}\right] \leq \binom{n}{k} (\mu/n)^k$ with k-wise independence

Chernoff Bound With Limited Independency

Save randomness, save money

- **Chernoff bound**: Let $X_1, \ldots, X_n \in \{0,1\}$ be ***k*-wise independent** trials

$$X = \sum_{i=1}^{n} X_i \quad \text{and} \quad \mu = \mathbb{E}[X]$$

- For any $\delta > 0$, $k \leq \dfrac{\mu\delta}{1 - \mu/n}$,

$$\Pr(X \geq (1+\delta)\mu) \leq \frac{\binom{n}{k}(\mu/n)^k}{\binom{(1+\delta)\mu}{k}} = \left(\frac{\mu}{n}\right)^k \cdot \prod_{i=0}^{k-1} \frac{n-i}{(1+\delta)\mu - i}$$

Is universal hashing a good notion?

At what cost we leverage independence?

- Time cost for evaluating k-universal hashing: $O(k)$
- Space cost: $O(k)$
- How much independence do we need?
  - Chaining / Maximum load: $\Theta(\log n / \log \log n)$
  - Linear probing: ≤ 5 [PR'07], ≥ 5 [PT'10]
  - Cuckoo hashing: $O(\log n)$
  - F_2 estimation: 4 [AMS'99]
  - min-wise independence (Jaccard similarity $J(A, B) = |A \cap B| / |A \cup B|$): $O(\log(1/\epsilon))$ [Indyk'00], $\Omega(\log(1/\epsilon))$ [PT'10]

Is universal hashing a good notion?

At what cost we leverage independence?

- Time cost for evaluating k-universal hashing: $O(k)$
- Space cost: $O(k)$
- Can we do better?
  - Not really [Siegel'89]:
    - $t < k$ time requires $|U|^{1/t}$ space;
    - $\Omega(1)$-universality in $O(1)$ time requires $|U|^{\Omega(1)}$ space
- Why?
  - Slow evaluating both in theory and in practice

But we care about only practice

A fast hashing in practice?

Tabulation hashing [Zobrist'79]

Bitwise operation is fast

- A hashing function based on bitwise operation?
- Decompose $x = (x_1, \ldots, x_c) \in [u]^c$
- c truly random hash table: $T_1, \ldots, T_c : [u] \to [m]$
- Hash value:

$$h(x) = T_1[x_1] \oplus \ldots \oplus T_c[x_c]$$

- Space cu, time $O(c)$
- Fits in cache for $c = 4$, $u = 256$, $u^c = 2^{32}$. Extremely fast in practice.

Tabulation hashing

Bitwise operation is fast

- Decompose $x = (x_1, \ldots, x_c) \in [u]^c$
- c truly random hash table: $T_1, \ldots, T_c : [u] \to [m]$
- Hash value:

$$h(x) = T_1[x_1] \oplus \ldots \oplus T_c[x_c]$$

- Space cu, time $O(c)$
- Tabulation hashing is 3-universal
- But is not even 4-universal: $h(a_1, a_2) \oplus h(a_1, b_2) \oplus h(b_1, a_2) \oplus h(b_1, b_2) = 0$

Tabulation hashing

Bitwise operation is fast

- Decompose $x = (x_1, \ldots, x_c) \in [u]^c$
- c truly random hash table: $T_1, \ldots, T_c : [u] \to [m]$
- Hash value: $h(x) = T_1[x_1] \oplus \ldots \oplus T_c[x_c]$ is not even 4-universal
- **But** works for all the aforementioned applications:
  - Chaining / Maximum load: $\Theta(\log n / \log \log n)$
  - Linear probing: ≤ 5 [PR'07], ≥ 5 [PT'10]
  - Cuckoo hashing: $O(\log n)$
  - F_2 estimation: 4 [AMS'99]
  - min-wise independence: $O(\log(1/\epsilon))$ [Indyk'00], $\Omega(\log(1/\epsilon))$ [PT'10]

Tabulation hashing

- Decompose $x = (x_1, \ldots, x_c) \in [u]^c$
- c truly random hash table: $T_1, \ldots, T_c : [u] \to [m]$
- Hash value: $h(x) = T_1[x_1] \oplus \ldots \oplus T_c[x_c]$ is not even 4-universal
- Most probabilistic analysis is based on independence
  - When is $h(x)$ independent of $h(y_1), h(y_2), \ldots$?
  - Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$
  - A set S is mutually independent if $\forall x \in S,\ \exists$ good position character

Peelability

First attempt

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S,\ \exists$ good position character
- **Claim**: For any $|S| = d$, $\exists$ peelable subset of size $t \geq \max\{d^{1/c}, \log_2 d\}$
- $n \leq m^{1-\epsilon}$ balls into m bins
  - Any set $|S| \geq d$ lands into a common bin w.p. $1/m^{t-1}$
  - $\binom{n}{t} < n^t$ possible sets. Max load by union bound: w.p. $n^t/m^{t-1} \leq m^{1-\epsilon t}$

Peelability

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S,\ \exists$ good position character
- **Claim**: For any $|S| = d$, $\exists$ peelable subset of size $t \geq \max\{d^{1/c}, \log_2 d\}$
- $t \geq d^{1/c}$
  - $S \subseteq S_1 \times S_2 \times \ldots \times S_c$, there is $i \in [c]$ s.t. $|S_i| \geq |S|^{1/c}$
- $t \geq \log_2 d$
  - Recursion: as long as $|S| > 1$, we can pick from $|S_i| > 1$

Peeling

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S,\ \exists$ good position character
- Idea:
  - Find a good PC α, then S_α is independent of $S_{-\alpha}$
  - Keep finding, get independent sets $S_\alpha, S_\beta, \ldots$
  - Apply peelablility argument on S_α, S_β respectively
  - Improvement overall

Anything wrong?

Peeling

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S,\ \exists$ good position character
- Idea:
  - Find a good PC α, then S_α is independent of $S_{_\alpha}$
  - Keep finding, get **non**-independent sets $S_\alpha, S_\beta, \ldots$
    - $\mathbb{E}[X_\alpha \mid X_\beta = x_\beta, \ldots] = \mathbb{E}[X_\alpha]$

A closer look at Chernoff (assume $x_i \in [0,1]$)

- We are examining in fact (choose $t = \ln(1+\delta)$)

$$\Pr[X \geq (1+\delta)\mu] \leq \Pr\left[e^{tX} \geq e^{t(1+\delta)\mu}\right] = \Pr\left[(1+\delta)^X \geq (1+\delta)^{(1+\delta)\mu}\right]$$

- $\mathbb{E}\left[(1+\delta)^X\right] = \sum_{x_{<n}} \left(\Pr[X_{<n} = x_{<n}] \cdot \mathbb{E}\left[(1+\delta)^X \mid X_{<n} = x_{<n}\right]\right)$

$$= \sum_{x_{<n}} \left(\Pr[X_{<n} = x_{<n}] \cdot (1+\delta)^{\sum_{i<n} x_i} \cdot \mathbb{E}\left[(1+\delta)^{X_n} \mid X_{<n} = x_{<n}\right]\right)$$

- $\mathbb{E}\left[(1+\delta)^Y\right] \leq \mathbb{E}[Y] \cdot (1+\delta)^1 + (1 - \mathbb{E}[Y]) \cdot (1+\delta)^0 = 1 + \delta\mathbb{E}[Y] \leq e^{\delta\mathbb{E}[Y]}$

- $\mathbb{E}\left[(1+\delta)^X\right] \leq \sum_{x_{<n}} \left(\Pr[X_{<n} = x_{<n}] \cdot (1+\delta)^{\sum_{i<n} x_i} \cdot e^{\delta\mathbb{E}[X_n]}\right) \leq e^{\delta\mathbb{E}[X]}$

Peeling

Chernoff: For $X_1, \ldots \in [0,d]$ with fixed means,

$$\Pr\left[\sum X_i \geq (1+\delta)\mu\right] \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu/d}$$

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S,\ \exists$ good position character
- Idea:
  - Find a good PC α, then S_α is independent of $S_{-\alpha}$
  - Keep finding, get **non**-independent sets $S_\alpha, S_\beta, \ldots$
    - $\mathbb{E}[X_\alpha \mid X_\beta = x_\beta, \ldots] = \mathbb{E}[X_\alpha]$
    - Upper bound? maximum load!

Peeling

Chernoff: For $X_1, \ldots \in [0,d]$ with fixed means,

$$\Pr\left[\sum X_i \geq (1+\delta)\mu\right] \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu/d}$$

- Position character (i, x_i) such that $\forall j, y_{j,i} \neq x_i$ makes $h(x)$ independent
- A set S is mutually independent if $\forall x \in S$, $\exists$ good position character
  - Find a good PC α, then S_α is independent of S_α **and** $|S_\alpha| \leq n^{1-1/c}$
  - Keep finding, get **non**-independent sets $S_\alpha, S_\beta, \ldots$ **with fixed means**
    - Max-load: for any bin, d balls from S_α land w.p. $m^{1-t/c}$
    - w.p. $1 - n \cdot m^{1-t/c}$, $X_\alpha \leq d$ for all α

max-load: $n \leq m^{1-\epsilon}$ balls into m bins,
max-load d w.p. $\leq m^{1-\epsilon t}$